

Microplastic Polymer Analysis Report

Report ID	afbea2be-473d-4a8b-b14a-4480bd475708
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Sample ID	FTIR-6855

Prediction Results

Detected Polymer	PP
Confidence Score	0.67%
Model Used	base_model

FTIR Spectral Analysis

- 1455 cm⁻¹ → CH bending
- 1375 cm⁻¹ → CH₃ symmetric deformation

Confidence Interpretation

■■■■■■■■■■ 0.67%

Scientific References

- [1] Karian, H. Handbook of Polypropylene.
- [2] Maier, C. Polypropylene: The Definitive User's Guide.
- [3] Andrady, A. Plastics and environmental sustainability.
- [4] Fotopoulou, K. Microplastics in marine systems.
- [5] Zhang, J. FTIR identification of polypropylene.

AI Generated Summary

Microplastic sample FTIR-6855 was analyzed using the *base_model* classifier, identifying the specimen as polypropylene (PP) with a confidence score of 0.67%. While PP is a widespread environmental contaminant, this low confidence value underscores the necessity for supplementary spectral validation to definitively confirm the polymer's identity. Improving the reliability of such automated identification models remains critical for accurate source-tracking and ecological risk assessments of microplastic debris.